🎓 Final Project — Customer Churn Prediction

🧠 Problem Statement

A company wants to understand **customer behavior** and predict:

> Will a customer leave the company (churn) or stay?

Customer churn is a major business problem because keeping existing customers is cheaper than acquiring new ones.

Your task is to use **data analysis, visualization, and machine learning** to help the company make better decisions.

📊 Dataset Overview

You are given a dataset with the following columns:

- **Monthly_Usage**

  Customer usage per month

- **Customer_Support_Calls**

  Number of times the customer contacted support

- **Subscription_Length**

  Number of months the customer has been with the company

- **Payment_Delay**

  Number of days the customer delays payment

- **Churn**
  - `0` → Customer stays

    - `1` → Customer leaves

📁 Dataset file: `customer_churn_dataset.csv`

🎯 Project Objectives

By the end of this project, you should be able to:

- Explore and understand a real dataset
- Visualize relationships between features and churn
- Train a **classification model**
- Evaluate model performance
- Explain results using **data storytelling**

🧪 Steps of Your Work

1️⃣ Data Exploration

- Load the dataset using Pandas
- Display:
    - `head()`
    - `info()`
    - `describe()`
- Check if the data is clean

2️⃣ Data Visualization

Create **at least 3 visualizations**, for example:

- Monthly_Usage vs Churn (scatter plot)
- Customer_Support_Calls vs Churn

- Bar chart showing Churn distribution

✔️ Add:

- Titles
- Axis labels
- Legends
- Proper figure size

3️⃣ Prepare Data for Machine Learning

- Define features (**X**)
- Define target (**y = Churn**)
- Split data into training and testing sets

4️⃣ Train a Classification Model

- Use **Logistic Regression**
- Train the model using training data

5️⃣ Model Evaluation

- Calculate **accuracy**
- Display **confusion matrix**
- Explain what the results mean

6️⃣ Predictions

- Predict churn for **new customers**
- Explain why the model gives these predictions

7️⃣ Interpretation & Storytelling

Write **4–6 sentences** answering:

- Which customers are more likely to leave?
- Which feature seems most important?
- Can we trust the model 100%? Why?
- How can the company reduce churn?

🎤 Presentation Requirements

You must prepare a **presentation (PowerPoint / Google Slides)** that includes:

Slide 1 — Title

- Project title
- Your name(s)

Slide 2 — Problem Description

- Explain the churn problem
- Why it matters for a company

Slide 3 — Dataset Overview

- Dataset size
- Columns description
- What each feature means

Slide 4 — Data Exploration

- Key observations
- Any interesting patterns

Slide 5 — Visualizations

- Show graphs
- Explain what they reveal

Slide 6 — Machine Learning Model

- Model used (Logistic Regression)
- Features used
- Why classification

Slide 7 — Results

- Accuracy
- Confusion matrix (bonus)
- Interpretation

Slide 8 — Predictions

- Example predictions
- Explanation

Slide 9 — Conclusion

- Key insights
- Business recommendations

📦 Final Deliverables

You must submit:

- ✅ Notebook (`.ipynb`)
- ✅ Presentation slides